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The Landscape Resilience Index: a flexible tool for integrated landscape management

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Globally, landscapes are subject to intensifying and interacting environmental pressures, making landscape resilience increasingly important for sustaining ecosystem services, livelihoods, and long-term development. Integrated landscape management offers a promising approach to strengthening resilience, yet many existing monitoring frameworks are not fit for purpose because they are too narrow in either scope or scale. This paper presents the Landscape Resilience Index (LRI), a flexible tool that integrates diverse primary and secondary datasets to monitor progress towards resilience over time across three interconnected pillars: land, water, and adaptive capacity. The LRI combines monitoring at site and landscape scales while aligning outcomes with national policy objectives. Using the Kilombero landscape in Morogoro, Tanzania, as a case study, we demonstrate how the LRI can provide a coherent picture of landscape resilience and identify priority interventions linked to national policy commitments. The framework strengthens the evidence base for adaptive management, policy alignment, and investment by linking local interventions to broader resilience outcomes and supporting more coordinated, scalable landscape action.

KEYWORDS

integrated landscape management, landscape resilience, nature-based solutions, restoration, sustainable land management

1 Introduction

Landscapes can be defined as socio-ecological systems comprising natural and human-modified ecosystems shaped by interactions amongst ecological, historical, economic, and socio-cultural processes (Denier et al., 2015). In this regard, landscape resilience refers to the capacity of a landscape to absorb disturbances, adapt to change, and continue to provide essential ecosystem services that sustain human wellbeing (1000 Landscapes for 1 Billion People, 2022). Strengthening landscape resilience is essential for sustaining agricultural productivity, reducing vulnerability to climate change, maintaining and enhancing ecosystems and their services, and supporting long-term economic and social development. Approximately 30% of people living in tropical countries depend directly on nature to meet their basic needs (Fedele et al., 2021). Yet, competing demands for water resources, carbon sequestration, agriculture, urban expansion, and energy production are intensifying land and water scarcity (King et al., 2023). These pressures coincide with accelerating biodiversity loss. Specifically, the Living

Planet Report (WWE, 2024) estimates declines of 85% in freshwater, 69% in terrestrial, and 56% in marine vertebrate populations between 1970 and 2020, highlighting the urgency of protecting ecosystems that underpin human wellbeing and resilience.

Conservation planning and environmental management under competing ecological and socio-economic objectives, particularly in developing countries, is highly complex and requires approaches capable of balancing multiple and often conflicting priorities. Integrated decision-making frameworks that bring together environmental, social, and economic considerations are increasingly being recognised as important for supporting decision-making for resource management and conservation planning (Karadeniz et al., 2026a). Given the complexity of these competing land uses, integrated landscape management (ILM) is widely promoted as an approach to balance environmental, social, and economic objectives to enhance landscape resilience. ILM supports landscape resilience because it explicitly recognises the interdependence of ecological, hydrological, and socio-economic processes and requires stakeholders to develop a shared vision and strategy for ecosystem stewardship (Reed et al., 2016). However, existing tools for monitoring landscape resilience often emphasise specific dimensions rather than providing integrated assessments across ecological, hydrological, and socio-economic domains. For example, the Ecosystem Resilience Index (Johnson et al., 2025) focuses on ecosystem structure and function, particularly forest resilience to wildfire, but does not incorporate socio-economic dimensions. The Spatial Resilience Index (Harwood et al., 2022) evaluates biodiversity persistence under climate change using spatial modelling but remains centred on ecological responses. Similarly, the Land Health Monitoring Framework (López, 2023) was developed to assess and monitor biodiversity across agroecosystems at multiple scales, but focuses primarily on belowground diversity, above-ground diversity, and habitat diversity, with limited consideration of hydrological and socio-economic dimensions. Other approaches, such as the Open Resilience Index (Feldmeyer et al., 2021), integrate social and environmental indicators but primarily target climate risk and vulnerable populations. The Landscape Resilience Framework (Beller et al., 2015) offers conceptual guidance on maintaining biodiversity and ecological functioning but does not provide an operational monitoring index. The Indicators of Resilience in Socio-Ecological Production Landscapes and Seascapes (SEPLS) under the Satoyama Initiative provide valuable, community-driven insights into socio-ecological resilience, but rely largely on perception-based scoring exercises and lack a dedicated, measurable focus on key landscape processes such as hydrology (UNU-IAS, Alliance of Biodiversity International and CIAT, UNDP GEF-SGP and IGES, 2024). The limited incorporation of hydrological indicators within many resilience frameworks is particularly concerning since water availability, storage, and catchment-scale flows are fundamental determinants of landscape resilience (Falkenmark et al., 2019). Frameworks that overlook hydrological dynamics risk constraining integrated landscape management from the outset by providing only a partial representation of system dynamics. This limitation is significant given that more than two billion people currently lack access to safe drinking water and projections suggest a potential 40% deficit in global freshwater supply by 2030 (Global Commission on the Economics of Water, 2023). Adopting a multi-dimensional perspective enables landscape interventions to be prioritised according to system-wide needs and trade-offs, supporting more effective and targeted actions. Recent advances in conservation planning show the value of more integrative

approaches, such as spatially explicit decision-support frameworks that link biodiversity, ecosystem services, and land-use planning to support both conservation and sustainable development objectives (Karadeniz et al., 2026b). Such approaches allow for the evaluation of trade-offs across socio-ecological systems and provide a stronger basis for evidence-informed planning and adaptive management.

Since adaptation to change is a defining characteristic of landscape resilience, frameworks designed to monitor resilience must themselves be flexible to remain relevant over time and across diverse landscape contexts. Effective landscape resilience frameworks therefore need to be capable of integrating diverse datasets to balance scientific robustness with the ability to adapt to different ecological, social, and institutional contexts. Quantitative methods including remote sensing and spatial modelling can provide scalable and consistent datasets, but they often underrepresent social processes, governance structures, and adaptive capacity (Schatte and Meyer, 2025). Conversely, participatory and perception-based approaches can capture critical local knowledge yet may lack consistency or comparability across sites (UNU-IAS, Alliance of Biodiversity International and CIAT, UNDP GEF-SGP and IGES, 2024). This flexibility is especially important because many landscape initiatives continue to be delivered through short, project-based cycles during which programmes may pause or close, and priorities can shift between funding windows. From a practitioner's perspective, frameworks that integrate diverse datasets reduce the costs of data collection, improving the financial feasibility of long-term monitoring. Without such flexibility, resilience assessments risk becoming costly and difficult to sustain, ultimately limiting their usefulness for adaptive management and investment prioritisation.

Monitoring frameworks must also be sufficiently transferable to link local land and water management actions with broader policy objectives at landscape, national, and international scales (Bailey and Buck, 2016). This is particularly important for catalysing large-scale investment and collaboration from both public and private actors, who increasingly seek evidence that interventions align with national policy targets and deliver measurable contributions to climate adaptation, biodiversity recovery, and sustainable development goals (Löfqvist et al., 2023; Zu Ermgassen and Löfqvist, 2024). However, many existing resilience assessment frameworks are implemented either at the site or community level, or at national and global scales, with limited capacity to explicitly connect local interventions to landscape-level processes or policy commitments. This fragmentation constrains the ability of practitioners, policymakers, and public or private investors to trace how on-the-ground actions contribute to progress against broader environmental and development objectives. These limitations are particularly evident in the context of Nature-based Solutions (NbS), where investment requires credible evidence of cumulative impacts and risk reduction. Consequently, while such frameworks generate important local insights, they remain limited in their ability to support cross-scale governance, comprehensive landscape management, or large-scale investment in nature-based solutions.

Taken together, existing landscape resilience frameworks lack the flexibility, hydrological integration, and cross-scale coherence needed to translate local land management interventions into policy-relevant evidence and investable outcomes for nature-based solutions. This paper proposes the Landscape Resilience Index (LRI) as a multi-dimensional, multi-scalar tool designed to provide a holistic approach to monitoring landscape resilience and informing decisions in integrated landscape management.

2 Context setting in which innovation occurs

The Landscape Resilience Index was developed and applied within the Sustainability and Inclusion Strategy for Growth Corridors in Africa initiative (SUSTAIN-Africa), led by the International Union for Conservation of Nature (IUCN). The initiative promotes integrated water and land management across rural landscapes in Tanzania and Mozambique (Figure 1), using NbS to strengthen the links between ecosystems, livelihoods, and agricultural production. These approaches are particularly relevant in African landscapes facing climate change, food and water insecurity, deforestation, and environmental degradation (Ejim-Eze, 2025).

SUSTAIN-Africa currently operates through two complementary programmes: SUSTAIN Pro and SUSTAIN Eco. SUSTAIN Pro focuses on developing sustainable food systems and productive landscapes in Tanzania and Mozambique, with activities including climate-smart agriculture, agroforestry, riverine restoration, and improved agricultural practices. In Tanzania, interventions are concentrated in the Kilombero and Ihemi landscapes, while in Mozambique they are implemented in the Vanduzi and Barué districts of Manica Province. SUSTAIN Eco operates in Tanzania and focuses on strengthening ecosystem stewardship and governance. Activities include restoration initiatives, climate-smart agriculture, biodiversity conservation, and community-based monitoring. The

programme prioritises inclusive approaches that support women, youth, and vulnerable groups while promoting alternative livelihoods such as agroforestry and beekeeping.

The LRI was developed in response to practitioner needs identified through the IUCN-SUSTAIN programmes. It is intended as a practical monitoring and decision support tool for practitioners, including NGOs, local institutions and policy actors, working within productive landscapes. The tool assesses the resilience of communities, land and water resources and is designed to support baseline assessment and continuous monitoring. In doing so, the LRI aims to help users prioritise interventions, support adaptive management, mobilise investment, and align local action with national and global policy commitments.

3 Key programmatic elements

3.1 Developing the LRI

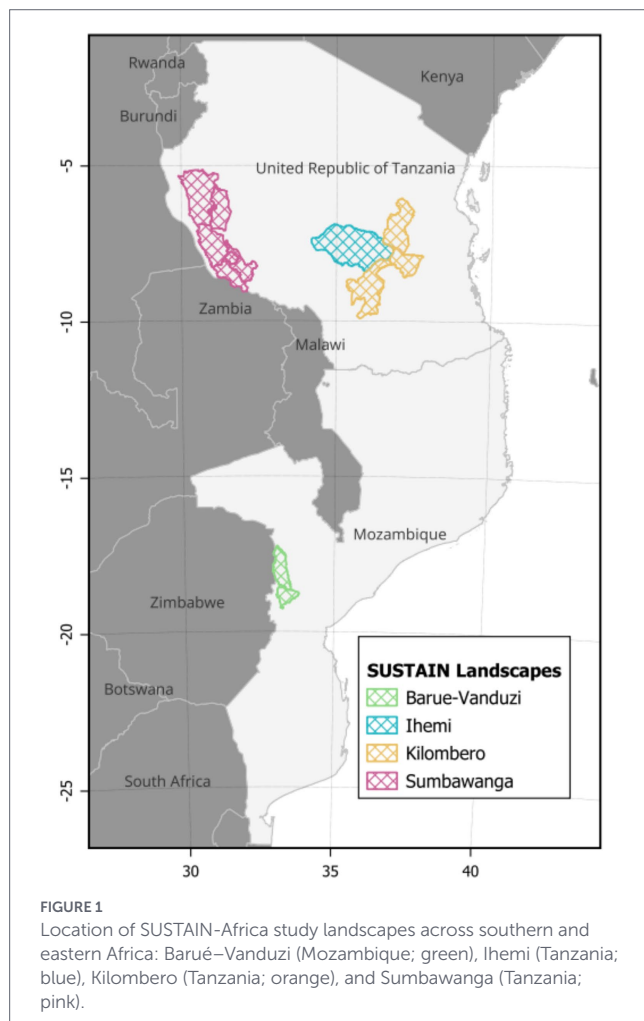
The LRI was developed specifically to address the need for a flexible, data-driven, and multi-dimensional landscape resilience monitoring tool which can be used to measure improvements and connect local action to national policy. As such the LRI is structured around three core principles.

3.1.1 Key principle 1. Integrating land, water, and adaptive capacity dimensions into one holistic index

The LRI asserts that landscape resilience frameworks must extend beyond single dimensions by explicitly integrating ecological, hydrological, and socio-economic dimensions into a single, structured framework. Consequently, the LRI is organised around three pillars: land, water, and adaptive capacity. This approach ensures that resilience is tracked not only in terms of environmental health but also in the capacity of communities and economic systems to adapt and sustain human well-being.

The pillars of the Landscape Resilience Index:

1. Land – Land indicators track both above- and below-ground biodiversity, carbon stocks, and soil fertility, ensuring that ecosystem health and productivity are captured in a manner that reflects the underlying ecological processes driving landscape resilience.
2. Water – Water indicators span site- and landscape-scale measures, including soil-water availability, water quality, water availability, and equitable access for multiple uses (drinking, irrigation, sanitation). This explicit inclusion of hydrological dynamics addresses a key gap in many resilience frameworks, acknowledging the centrality of water in sustaining ecological function and community livelihoods.
3. Adaptive capacity – Adaptive capacity indicators capture socio-economic dimensions, including agricultural and agroforestry productivity, adoption of sustainable land management practices, household poverty, perceptions of shocks and stresses such as flooding or drought events as well as land cover and landscape biodiversity.



By combining ecological, hydrological, and socio-economic measures within a single operational framework the LRI extends beyond current resilience monitoring tools that focus on individual system components. In doing so, it provides a coherent representation of landscape resilience that aligns local-scale actions with broader ecosystem and community outcomes.

3.1.2 Key principle 2. The LRI is relevant for stakeholders at different spatial scales

The LRI frames landscape resilience across three nested spatial scales (site, landscape, and national) enabling local interventions to be linked directly to national targets (e.g., Sustainable Development Goals, African Forest Landscape Restoration Initiative, Bonn Challenge). This provides a crucial mechanism for aligning grassroots efforts with government commitments, supporting more coordinated and evidence-based investment decisions (Table 1).

At the site level which focuses on soil and farms, the index monitors soil biological structures, farm system diversity (like crops and habitats), and soil-water interactions (including infiltration, moisture, and irrigation practices). This is the most detailed scale of assessment and generates data to guide farm-level decisions and advisory services. It typically involves on-the-ground data collection and laboratory-based analysis.

At the landscape level, the LRI captures broader ecological and hydrological processes that span across farms and administrative borders. Key indicators include catchment-level water availability and quality, ecological connectivity, use of sustainable land management practices, and exposure to extreme weather events. This level helps link individual actions to broader environmental outcomes which is essential for integrated landscape planning, ecosystem restoration, and climate resilience.

At the national level, the index connects local monitoring efforts to national policies and international goals. Although the national level is not a formal scale for direct monitoring, it plays a critical contextual role in supporting decision-making and understanding how actions in one productive landscape may impact resilience elsewhere. This broader perspective helps ensure that the framework accounts for ecological consequences that extend beyond the immediate target area.

By linking site- and landscape-level monitoring to national-level policy, the LRI provides multi-scale structure for connecting local interventions to national goals and commitments. In doing so, it supports landscape-level improvements in land and water management while helping governments and implementing organisations demonstrate alignment with national and global targets.

3.1.3 Key principle 3. The LRI is designed to adapt to varying landscape contexts

Indicator selection followed an iterative process designed to balance methodological robustness with practical applicability across different SUSTAIN landscapes. First, a long list of potential indicators was developed from a literature search, expert knowledge, and key informant interviews and organised into thematic groups. The indicators were then shortlisted to produce the LRI. Initially, indicator screening assumed that most data would require primary collection. Therefore, each indicator was assessed using a 2-stage strength

TABLE 1 Scales, pillars and indicators of the Landscape Resilience Index (LRI).

Scale	Pillar	Indicator
Site	Land	Above-ground biodiversity
		Below-ground biodiversity
		Carbon stocks
		Soil fertility
	Water	Soil-water availability
		Site level water biological activity
		Site level water chemistry
Landscape	Water	Landscape water access
		Landscape water availability
		Landscape water chemistry
	Adaptive capacity	Adoption of Sustainable Land Management (SLM) practices
		Agricultural, agroforestry and tree plantation productivity
		Extreme weather
		Household poverty
		Landcover percentage
		Landscape biodiversity
		Perception of shocks and stresses
National	Water	Water quality and quantity
		Access to quality water for different uses: drinking water, sanitation, irrigation
	Adaptive capacity	Agricultural productivity
		Carbon stored in forest and agricultural land including associated emissions reductions
		Adoption of SLM practices, certification and agroecological markets
		Healthy and diverse ecosystems
		Resilient communities to shocks, disasters and climate change
		Poverty rate

measure. First, the indicators were scored according to their feasibility and scalability, including cost, technical expertise, evidence availability, data precision, and whether the indicator could inform site, landscape, and national-level monitoring. Second, the indicators were scored according to their alignment with ongoing interventions, links to ecosystem service categories, and the number of wider co-benefits captured by the indicator. Each criterion was scored on a 0–3 scale, with higher scores indicating greater feasibility, stronger evidence, higher data precision, broader scalability, or greater relevance to ongoing activities and ecosystem-service outcomes (Supplementary Table 1).

The LRI was then validated through a participatory workshop with local stakeholders from the Kilombero and Ihemi landscape. During this process, it became evident that the landscapes experience different pressures on resilience and that data availability varied considerably. To account for these differences in landscape contexts and data availability, the LRI was revised so that the thematic groups would constitute the main LRI indicators, while

specific units of measurements would be used as sub-indicators. This structure allowed the LRI to retain a consistent overall framework while remaining flexible enough to accommodate differences in data availability, landscape priorities, and local challenges as the tool was scaled from Kilombero and Ihemi to Sumbawanga in Tanzania and Barué-Vanduzi in Mozambique. For example, the soil fertility indicator is consistently monitored across landscapes, but the specific datasets used may differ. In Kilombero, the soil fertility indicator is represented using secondary datasets on soil carbon to nitrogen ratio, soil pH, soil total nitrogen and soil phosphorus. Whereas in Barue-Vanduzi, the soil fertility indicator is represented by soil electrical conductivity and soil pH, since these datasets were readily available. Where data availability allowed, for some indicators the land use in addition to the data type was used as a sub-indicator. For example, the above-ground biodiversity indicator in Kilombero was comprised of bird species richness, mammal species richness and arthropod species richness in forest, cropland and grasslands. Where secondary data was unavailable, such as for soil fertility data in Ihemi and Sumbawanga landscapes, targeted primary data collection was conducted to ensure that each indicator is represented by at least one measurable sub-indicator within each landscape. The selected sub-indicators for primary data collection were based on the strength measure used on the original LRI.

For the national level indicators, a policy review was conducted for Tanzania and Mozambique to identify national targets, commitments, and reporting frameworks relevant to the LRI indicators. These were grouped under the corresponding national-level LRI indicators, with individual policies, targets, or commitments treated as sub-indicators. This allowed the LRI to connect landscape-level monitoring to national policy priorities while recognising that relevant commitments differ between countries.

3.2 Indicator scoring

For monitoring tools to support decision-making, information must be scientifically credible, but also salient, legitimate, and usable for the intended users (Cash et al., 2003; McNie, 2007). In practice, this means that monitoring frameworks must be practically feasible and sufficiently interpretable to support communication and inform management decisions (van Oudenhoven et al., 2018; Czúcz et al., 2021; Sedgwick et al., 2021). Since the LRI is intended for use by diverse stakeholders, including farmers, extension officers, Water User Association members, local practitioners, and government officials, we adopted a four-point scoring system: poor (1), moderate (2), good (3), and excellent (4). Thresholds for classification were defined using a combination of scientific literature and, where available, locally relevant policy standards (Table 2). In cases where neither literature nor policy or programme benchmarks were available, expert judgement was used to determine appropriate scoring thresholds (Supplementary Table 2).

Scores were applied at the sub-indicator level using the spatial scale appropriate to each data set. Scores were subsequently aggregated to derive indicator-level scores. For example, site-level measurements were scored at the sampling/site scale, spatial datasets such as land cover were scored at the mapped landscape or cluster scale and landscape water-availability indicators were scored at the selected river-reach scale. For indicators composed of multiple

sub-indicators (e.g., soil fertility), individual sub-indicator scores were equally weighted for aggregation to the indicator level. Equal weighting was used as a default approach because no empirically derived, policy-mandated, or stakeholder-agreed weighting scheme was available for the Kilombero baseline assessment. This standardised scoring system enabled consistent evaluation of diverse datasets while maintaining comparability across indicators.

3.3 Prioritising interventions

Following indicator scoring, baseline results were combined with the policy review to identify intervention priorities (Figure 2). This allowed lower-scoring indicators to be interpreted in relation to both local landscape needs and relevant national targets, commitments, and reporting frameworks.

3.4 Case study: Kilombero landscape

Here, we demonstrate how the Landscape Resilience Index can be applied to conduct a baseline assessment of landscape resilience and identify priority interventions aligned to national policy, using the Kilombero landscape in Tanzania as a case study.

3.4.1 Study area

The SUSTAIN-Africa Kilombero landscape covers the districts of Kilombero, Ifakara, Kilosa, Malinyi and Mlimba within the Southern Agricultural Growth Corridor of Tanzania (SAGCOT). It encompasses part of the Julius Nyerere National Park, Udzungwa Mountains National Park, the Kilombero Valley Floodplain, Kilombero RAMSAR site, Kilombero Forest Nature Reserve, Udzungwa Scarp Nature Reserve as well as the Mikumi National Park borders (Figure 3). The Kilombero landscape is recognised for both its high biodiversity (Myers et al., 2000) and agricultural productivity (Kangalawe and Liwenga, 2005). This landscape therefore represents a complex socio-ecological system in which ecological condition, water resources, agricultural production, and livelihoods are closely interconnected. In the Kilombero case study, resilience is assessed in relation to both natural resources and livelihood activities. In this regard, agricultural expansion, shifting land-use patterns, ecosystem degradation, water-management pressures (Malley et al., 2023; Msofe et al., 2024), and climate-related shocks represent key pressures on Kilombero's landscape resilience. Since these pressures affect the landscape at different temporal scales, the LRI could support an initial baseline assessment of these conditions, followed by repeat monitoring to evaluate whether intervention priorities and landscape-management responses are contributing to progress towards resilience over time.

SUSTAIN facilitates the coordination of risk responses through an Integrated Landscape Management (ILM) approach, enabling governance and planning to take place across stakeholders, sectors, and land uses. Sustainable Land Management (SLM) practices, including crop rotation, organic compost and fertiliser use, integrated pest management, terracing, erosion control, agroforestry, and improved soil-water management, operate as a risk-mitigation pathway by reducing pressures on land, water, production systems, and livelihoods. At the same time, Nature-based Solutions (NbS) provide an opportunity-creation pathway by protecting, restoring, and

TABLE 2 Summary of sub-indicators, data sources, threshold sources, and threshold basis used in the Landscape Resilience Index.

Indicator	No. sub-indicators	Sub-indicator	Sub-indicator details and data source	Threshold basis
Land pillar (site-level only)				
Above-ground biodiversity	11	Bird species richness	Sugarcane plantation, Smallholder farms, forest, grassland (Davis et al., 2024 ¹)	Expert—/programme-informed provisional thresholds
		Athropoda spp. richness	Cropland, Forest, Grassland, Woodland (Pfeifer, 2024 ¹)	Expert—/programme-informed provisional thresholds
		Mammal spp. richness	Cropland, Forest, Grassland (Pfeifer, 2024 ¹)	Expert—/programme-informed provisional thresholds
Below-ground biodiversity	3	Earthworm count	Cropland, forest, grassland (Pfeifer, 2024 ¹)	Literature-based thresholds (Phillips et al., 2019)
Carbon stocks	8	Above-ground Carbon	Cropland (Muthuri et al., 2023 ²); Forest (lowland and montane) (Mwambala et al., 2019 ³); Grassland (Laswai, 2011 ²); Woodland (Shirima et al., 2011 ²)	Literature-based thresholds (Mauya et al., 2019)
		Soil Organic Carbon	Cropland, Forest (lowland and montane), Grassland, Woodland (Pfeifer, 2024 ¹)	Policy linked thresholds- cropland (Changwon Initiative); Literature-based thresholds- forest, grassland, woodland (Mauya et al., 2022)
Soil fertility	4	Soil C: N	Cropland (Ovenden, 2025 ¹)	Literature-based thresholds (Brust, 2019)
		Soil pH	Cropland (Pfeifer, 2024 ¹)	Expert—/programme-informed provisional thresholds
		Soil total N	Cropland (Pfeifer, 2024 ¹)	Literature-based thresholds (Sirappa and Tandisau, 2018)
		Soil Phosphorus	Cropland (Pfeifer, 2024 ¹)	Literature-based thresholds (Heckman, 1998)
Water pillar (site-level)				
Soil-water availability	1	Plant available water content	% soils total volume is water available to plant in 0-20 cm layer (Ovenden, 2025 ¹)	Literature-based thresholds (Datta et al., 2017)
Site level biological activity of water	2	MiniSASS	Sandy river, Rocky river (MiniSASS, n.d. ¹)	Protocol-based thresholds (MiniSASS, n.d.)
Site level water chemistry	3	EC	— (Davis, 2026 ¹)	Literature-based thresholds (Kadhem, 2013)
		pH	— (Davis, 2026 ¹)	Literature-based thresholds (Scottish Environment Protection Agency, 2024)
		TDS	— (Davis, 2026 ¹)	Literature-based thresholds (Kadhem, 2013)
Water pillar (landscape-level)				
Landscape water access	1	Clean source of water for home consumption	Wells or Covered well, spring or tap (Ipsos Center for Development Research and Evaluation, 2023 ⁴)	Expert—/programme-informed provisional thresholds
Landscape water availability	4	River discharge	GEOGLOWS River discharge (Kilombero and Ruaha Rivers) in March and September (Hales et al., 2022)	Dataset—/reference-period-derived thresholds
Landscape level water chemistry	3	EC	— (Davis, 2026 ¹)	Literature-based thresholds (Kadhem, 2013)
		pH	— (Davis, 2026 ¹)	Literature-based thresholds (Scottish Environment Protection Agency, 2024)
		TDS	— (Davis, 2026 ¹)	Literature-based thresholds (Kadhem, 2013)

(Continued)

TABLE 2 (Continued)

Indicator	No. sub-indicators	Sub-indicator	Sub-indicator details and data source	Threshold basis
Adaptive capacity pillar (landscape-level only)				
Adoption of Sustainable Land Management (SLM) practices	5	% of sustainable land management practices	enhance soil fertility, Apply manure, Crop rotation, reduce soil erosion; add compost (Ipsos Center for Development Research and Evaluation, 2023 ⁴)	Expert—/programme-informed provisional thresholds
Agricultural, agroforestry and tree plantation productivity	6	Average crop yield	Rice, Sunflower, Soybean, Sugarcane (growers), Sugarcane (estate irrigated), Sugarcane (estate rainfed) (Metta, 2024 ⁵)	Expert—/programme-informed provisional thresholds
Extreme weather	4	Average percentage land area flooding in any 1 month	Kilombero, Kilosa (FAO EVE Global Flood Monitoring System ³)	Expert—/programme-informed provisional thresholds
		Percentage area affected by drought in 2024	Kilombero, Kilosa (Combined Drought Index (CDI) ³)	Expert—/programme-informed provisional thresholds
Household poverty	1	PPI	Percentage likelihood of households living below poverty line \$1.90 (Metta, 2024 ⁵)	Expert—/programme-informed provisional thresholds
Landcover percentage	1	% tree cover LOSS per annum	Reduction in tree cover compared to 2000 baseline (globalforestwatch.org ²)	Dataset—/reference-period-derived thresholds
Landscape biodiversity	1	Proportion of spp. expected to persist	Biodiversity Habitat Index (Harwood et al., 2022 ³)	Dataset—/reference-period-derived thresholds
Perception of shocks and stresses	2	% of people who report x as a big problem	Floods and storms, Drought or lack of rain (Ipsos Center for Development Research and Evaluation, 2023 ⁴)	Expert—/programme-informed provisional thresholds

Numerical codes indicate data provenance: ¹secondary field, laboratory, or protocol dataset; ²published secondary data or literature dataset; ³global/modelled spatial or hydrological dataset; ⁴survey or monitoring, evaluation and learning (MEL)-derived data; and ⁵programme monitoring or target data.

enhancing ecosystem functions through actions such as riparian restoration, buffer-zone creation, protection of sensitive areas, restoration of degraded land, and maintenance of ecological connectivity. Together, these approaches translate the concept of landscape resilience into practical action, while the LRI provides a monitoring and adaptive-management tool to assess progress towards resilience (Figure 4).

3.4.2 Sub-indicator data collection and scoring

The baseline assessment of landscape resilience in the Kilombero landscape was conducted exclusively using secondary data derived from publicly available spatially explicit studies, SUSTAIN-Africa datasets, and global databases. In total, 60 distinct sub-indicators were compiled to represent ecological, hydrological, and socio-economic dimensions of landscape resilience (Table 2).

Under the land pillar, the majority of the baseline data was derived from existing datasets. Above and below ground biodiversity datasets were derived from the Agrisys Tanzania project (Pfeifer, 2024). The carbon stocks dataset included literature reported values for cropland (Muthuri et al., 2023), forests (Mwambala et al., 2019), grassland (Laswai, 2011), and woodland (Shirima et al., 2011). Below-ground carbon were also obtained from the Agrisys Tanzania project covering cropland, forest, grassland and woodland (Pfeifer, 2024). The soil fertility indicator focused on croplands and covered soil C:N (Ovenden, 2025), soil pH, soil total nitrogen and soil phosphorus (Pfeifer, 2024). Thresholds were defined using a combination of literature

benchmarks, national datasets, and expert judgement (Supplementary Table 2). For example, thresholds for earthworm diversity were based on regional ranges from Phillips et al. (2019), while soil organic carbon and above-ground carbon thresholds were derived from national analyses of the NAFORMA datasets (Mauya et al., 2019; Mauya et al., 2022). Soil fertility thresholds were informed by published agronomic ranges (e.g., Brust, 2019; Sirappa and Tandisau, 2018; Heckman, 1998).

Indicators within the water pillar were compiled from multiple sources. The biological water activity indicator was comprised of macroinvertebrate assemblage data using the mini stream assessment scoring system (MiniSASS, n.d.). Here, the thresholds for classification for the LRI were derived from the MiniSASS ecological categories, with sandy and rocky substrate rivers assessed separately. Water chemistry data (both site and landscape level) was obtained from Davis (2026), with the distinction between river stretches used for site-level versus landscape level indicators determined from the stream's Strahler Order number (Strahler, 1957). Soil water availability data was taken from Ovenden (2025), with thresholds for classification for water electrical conductivity and turbidity based on recommended values for irrigation water in Kadhem (2013). Water pH was based on the environmental standards for clear waters defined by the Scottish Environment Protection Agency (2024). Landscape-scale water availability was assessed using the average daily river discharge data from the GEOGLOWS Hydrologic Model v2 dataset (Hales et al., 2022). GEOGLOWS v2 uses the ERA5/ERA5-Land runoff data to simulate streamflow along river networks and consists of over 7 million

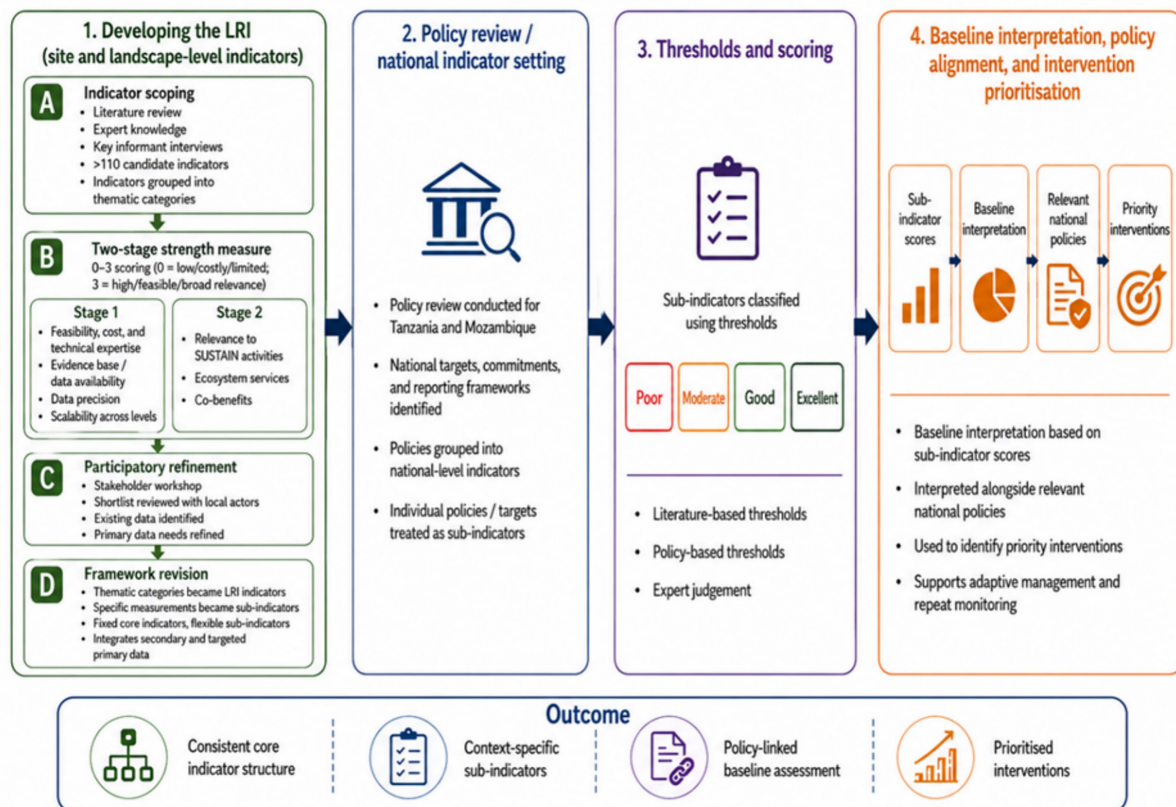


FIGURE 2

Development and application workflow of the Landscape Resilience Index (LRI). The figure summarises the main stages used to develop and apply the LRI. First, site- and landscape-level indicators were developed through indicator scoping, a two-stage strength measure, participatory refinement with stakeholders, identification of existing data sources, and subsequent framework revision. Second, a policy review was undertaken in Tanzania and Mozambique to define national-level indicators and associated policy sub-indicators. Third, thresholds were set for sub-indicators and used to classify condition on a four-point scale (poor, moderate, good, excellent), drawing on literature-based thresholds, policy-based thresholds, and expert judgement where required. Finally, sub-indicator scores were used to generate a baseline interpretation, which was then considered alongside relevant national policies to identify and prioritise landscape interventions.

river segments, globally. For the Kilombero case study, average daily discharge for one segment of the Great Ruaha River (GEOGLOWS ID: 110540262) and the Kilombero River (GEOGLOWS ID: 110500912) for March and September 2024 were used, representing the long rainy and long dry season, respectively. Percentage deviation above or below the 30-year average was used to determine the landscape-scale water availability score, with average daily discharge values closer to the long-term average receiving higher scores. The full threshold categories are provided in the [Supplementary Table 2](#).

In the adaptive capacity pillar, socio-economic indicators (adoption of sustainable land management practices, agricultural, agroforestry and tree plantation productivity, household poverty, and perception of shocks and stresses) primarily relied on survey data from SUSTAIN-Africa baseline reports. Here, thresholds were set based on discussion with key stakeholders. Land cover percentage used data from the Global Forest Watch ([Hansen et al., 2013](#)), with thresholds for classification based on historical tree cover loss for the landscape. Data for the landscape biodiversity indicator was obtained from the BHI v2: Biodiversity Habitat Index: 30s global time series dataset ([Harwood et al., 2022](#)), which assesses the proportion of species expected to persist over time. The extreme weather indicator data was taken from the Humanitarian Data Exchange, specifically the FAO EVE Global Flood Monitoring

System dataset¹ and the Intergovernmental Authority on Development Climate Prediction and Applications Centre's East Africa Drought Watch Combined Drought Index dataset.² For each of the extreme weather indicators, thresholds for classification were set based on discussion with SUSTAIN-Africa.

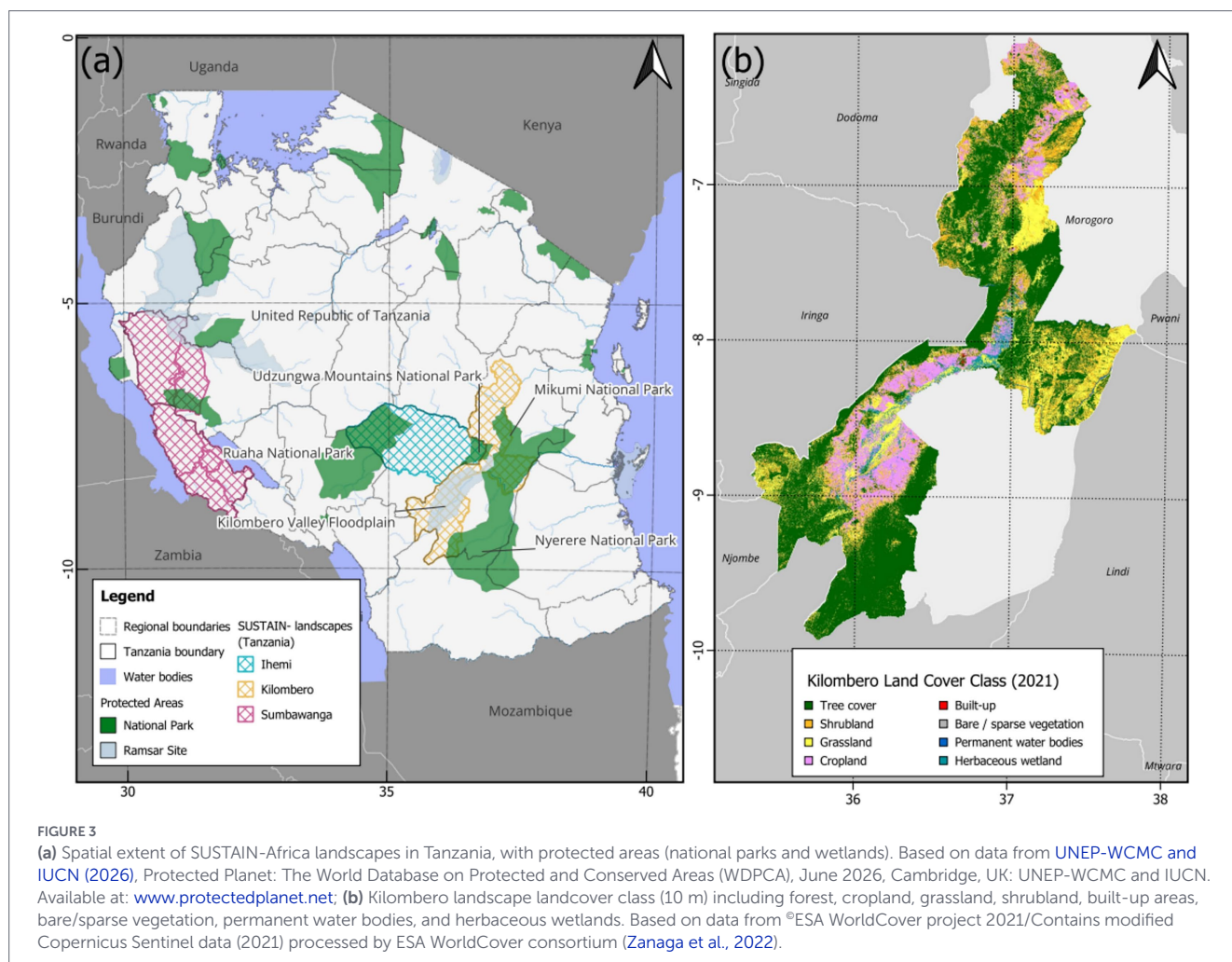
Quantitative changes in LRI indicator scores are intended to be assessed through repeat monitoring, where new sub-indicator values are compared against the same predefined threshold categories. As the Kilombero case study represents a baseline assessment, temporal trends cannot yet be inferred.

3.4.3 Baseline assessment findings

The baseline assessment for Kilombero ([Table 3](#)) indicates mixed performance across the three pillars of landscape resilience. Within the land pillar, above-ground biodiversity, below-ground biodiversity and carbon stocks were rated as good, whereas soil fertility was rated as moderate. In the water pillar, site level water chemistry and soil-water availability were rated as excellent, but small stream biological activity was good. Although landscape

¹ <https://data.humdata.org/dataset/fao-eve-global-flood-monitoring-system>

² <https://droughtwatch.icpac.net/drought-indicators/cdi/>



level water chemistry and water availability were rated as excellent and good, respectively, landscape water access was rated as moderate. For the adaptive capacity pillar, extreme weather was rated as excellent, whereas, household poverty, landscape biodiversity and perception of shocks and stresses were rated as good. The indicators agricultural, agroforestry, tree plantation productivity, adoption of SLM practices and landcover percentage were rated as moderate.

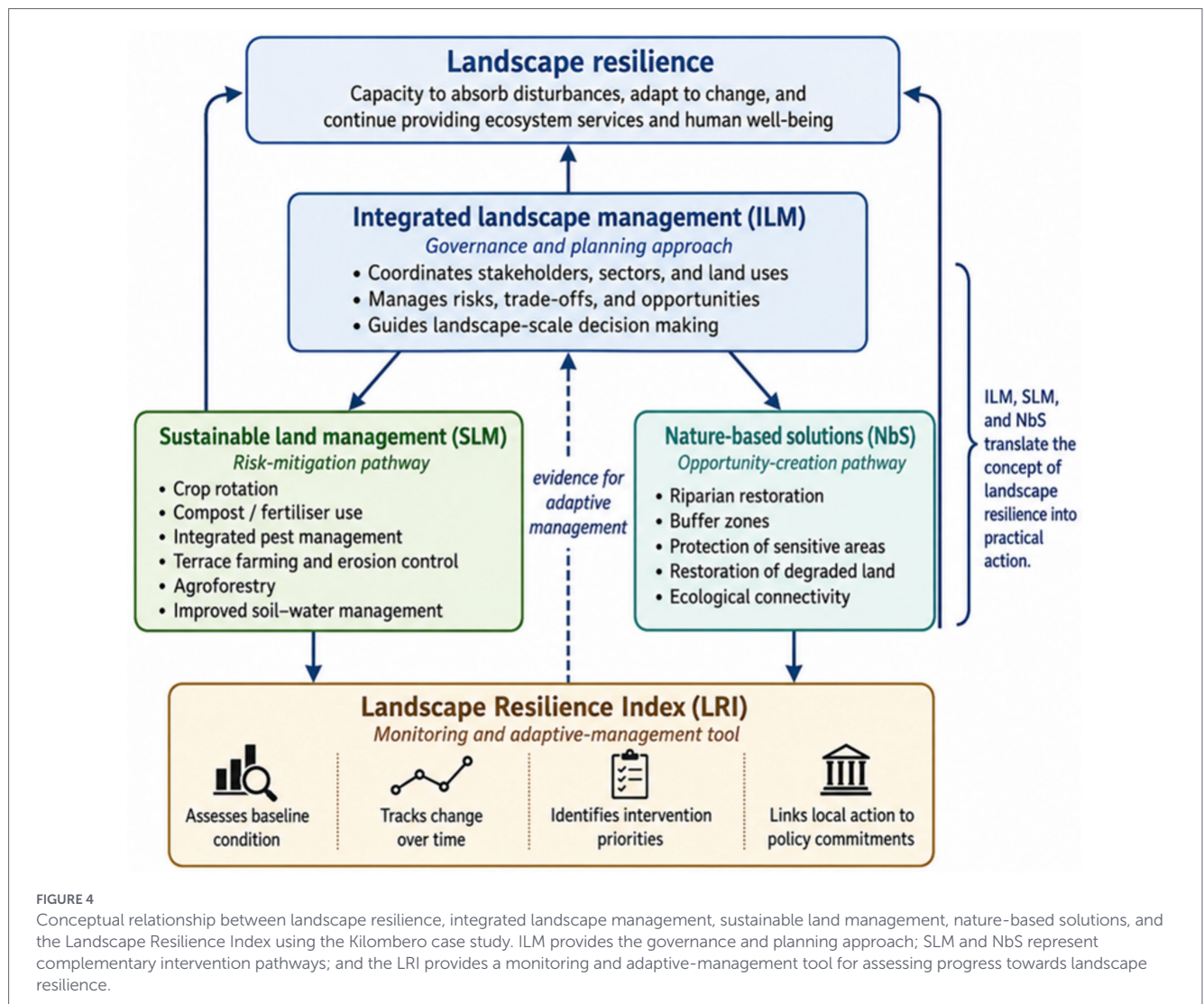
The baseline assessment also illustrates how several LRI indicators can be interpreted in relation to broader Tanzanian policy commitments. Specifically, indicators relating to water access, water quality, sustainable land management, ecosystem restoration, agricultural productivity, climate resilience, and poverty reduction correspond to several national targets and reporting frameworks currently relevant in Tanzania (Table 4).

3.4.4 Recommendations for the Kilombero landscape

Although the baseline assessment indicates that landscape resilience in the Kilombero landscape is generally good to excellent, two priority intervention areas emerge: the sustainable intensification of agriculture and improving access to water for multiple uses.

3.4.4.1 Sustainable intensification of agriculture

Sustainable intensification (SI) aims to increase agricultural yields without expanding cultivated land or causing adverse environmental impacts (Baulcombe et al., 2009). Within the LRI, the soil fertility indicator was therefore interpreted in relation to the capacity of cropland to support productive and sustainable agriculture. In Kilombero, soil fertility was classified as moderate, primarily because low soil total nitrogen, assessed against thresholds from Sirappa and Tandisau (2018), and low available phosphorus (Bray I method), assessed using thresholds from Heckman (1998), reduced the overall indicator score. This was accompanied by a slightly sub-optimal carbon-to-nitrogen (C:N) ratio (Brust, 2019), despite generally favourable soil pH. The moderate soil fertility score therefore suggests that nutrient availability, rather than soil acidity or alkalinity, is the main constraint to improving cropland condition. Phosphorus deficiency is a major constraint on crop production in sub-Saharan Africa (Nziguheba et al., 2016), with soil erosion being a key pathway for phosphorus loss (Carpenter and Bennett, 2011). The moderate C:N ratio which was slightly below the optimal range of 20–30:1 (Brust, 2019), further suggest limited carbon inputs relative to nitrogen; however given that total nitrogen is also low, the results more broadly indicate depleted soil nutrient stocks,



potentially associated with nutrient leaching, erosion, or low organic matter inputs. From a management perspective, this indicates that interventions should prioritise practices that improve nutrient retention, reduce soil loss, and increase organic matter input. Here, NbS interventions³ such as vegetative buffer strips (Apollonio et al., 2021), agroforestry (Awazi, 2025), conservation agriculture (Diop et al., 2022), and mulching (Bogunović and Filipović, 2023) may reduce soil erosion and nutrient leaching, thereby enhancing phosphorus retention and improving soil nitrogen-supplying capacity (Al-Kaisi et al., 2005).

At the landscape level, moderate adoption of sustainable land management practices (including practices to enhance soil fertility, reduce soil erosion, manure/compost application and crop rotation) indicates that farmers are likely receptive to such interventions, the success of which could be tracked through the site-level soil fertility indicator and the landscape level indicator of agricultural, agroforestry and tree plantation productivity.

These interventions support national agricultural productivity objectives under the SAGCOT Green Print and contribute to

emissions reduction and carbon storage targets by improving soil organic carbon, in line with the Changwon Initiative. At the landscape level, scaling up adoption of sustainable land management and agro-ecological practices strengthens ecosystem function, stabilises production, and reduces vulnerability to shocks. Increased uptake directly contributes to national commitments under the Tanzania National Ecological Organic Agriculture Strategy (2023–2030) and the National Climate Change Response Strategy (2021–2026), which targets widespread adoption of climate-smart land management.

3.4.4.2 Improving access to water for multiple uses

The second key priority for intervention is improving access to water for multiple uses. The landscape water access indicator, which measured access to water for home consumption, was rated moderate, with only 36% of survey respondents reporting access to a covered well, spring, or tap. Landscape-level investment in protected wells, managed water access points, and associated infrastructure would directly address this vulnerability, with progress measurable through the LRI's landscape water access indicator. At the national scale, improved access to safe drinking water would also contribute directly to SDG VNR 1.4.1, which tracks access to drinking water and sanitation, and support United Republic of Tanzania (2021) commitment to achieve universal access to clean and safe water in rural areas by 2030.

³ Nature-based Solutions implemented through IUCN-related interventions should be aligned with the eight criteria and 27 indicators of the second edition of the IUCN Global Standard for Nature-based Solutions™

TABLE 3 Indicator score and subsequent health category for Kilombero.

Pillar	Indicator	Average score	Landscape resilience category
Site			
Land	Above-ground biodiversity	2.64	Good
	Below-ground biodiversity	3.00	Good
	Carbon stocks	2.50	Good
	Soil fertility	2.00	Moderate
Water	Soil-water availability	4.00	Excellent
	Site level water biological activity	3.50	Good
	Site level water chemistry	4.00	Excellent
Landscape			
Water	Landscape water access	2.00	Moderate
	Landscape water availability	2.75	Good
	Landscape water chemistry	4.00	Excellent
Adaptive Capacity	Adoption of Sustainable Land Management (SLM) practices	2.00	Moderate
	Agricultural, agroforestry and tree plantation productivity	2.00	Moderate
	Extreme weather	3.50	Excellent
	Household poverty	3.00	Good
	Landcover percentage	2.00	Moderate
	Landscape biodiversity	3.00	Good
	Perception of shocks and stresses	2.50	Good

Landscape resilience categories are Poor: <1.50; Moderate: 1.50–2.49; Good: 2.50–3.49; Excellent: = > 3.50.

In addition to improving water access for home consumption, expanding small-scale irrigation infrastructure in the Kilombero landscape would likely improve crop productivity (Gebbru et al., 2025; Mupaso et al., 2023; Senbeta, 2023) and reduce sensitivity to rainfall variability and drought (Aguilar et al., 2022; Lankford et al., 2023). While irrigation water access was not captured in the Kilombero baseline assessment, incorporating this measure into future surveys would allow the impacts of water-related interventions to be assessed more comprehensively. These benefits could be tracked through the LRI's landscape water access indicator under the water pillar, as well as the agricultural, agroforestry, and tree plantation productivity indicator and the perception of shocks and stresses indicator under the adaptive capacity pillar. At the national scale, improved irrigation access would contribute to Tanzania's SDG VNR 2.4.1 goal by increasing the area under irrigation and supporting climate-resilient agricultural growth. Strengthening water monitoring and basin-level management would further support national objectives under SDG VNR 6.3.2 and the National Climate Change Response Strategy (2021–2026), which prioritises water resilience across all basins.

4 Discussion

Historically, investment in landscape resilience interventions has been constrained by a lack of coordinated evidence linking local conditions to landscape-level risks, co-benefits, and policy or value-chain priorities (Sayer et al., 2013; Shiri et al., 2025; Reed et al., 2016; Kusters et al., 2020). While various landscape and resilience assessment frameworks exist, their practical application is often constrained by limited

integration across land, water, and livelihood dimensions, and weak links between local monitoring, intervention planning, and policy reporting. These limitations can make it difficult for practitioners, governments, and investors to translate landscape-level evidence into coordinated management decisions.

The Kilombero case study demonstrates the practical contribution of the LRI by (1) integrating diverse land, water, and adaptive-capacity datasets into a structured baseline assessment; (2) supporting continued monitoring, reporting and verification to facilitate adaptive management; and (3) integrating on-the-ground monitoring into policy and governance processes.

First, the Kilombero case study shows how the LRI can provide a structured baseline of landscape resilience covering the three pillars of landscape resilience. In doing so, the LRI helps identify where conditions are relatively strong and where intervention may be needed (Reed et al., 2016; Milder et al., 2014). For example, the baseline assessment of Kilombero indicated mixed performance across the three pillars. Good scores for above-ground biodiversity, below-ground biodiversity, and carbon stocks suggest that important ecological functions remain present within the landscape, whereas the moderate soil fertility score indicates constraints on the capacity of cropland to support sustainable intensification, particularly where low nitrogen and phosphorus availability may limit crop production.

Second, the LRI supports continued monitoring, reporting, and verification to capture long-term trends and intervention impacts, which are increasingly required to demonstrate impact and ensure accountability (Chausson et al., 2020; Herold and Skutsch, 2011; Lindenmayer and Likens, 2010). In the Kilombero case study, the current scores should be interpreted as an initial baseline rather than evidence of temporal change. However, if repeated using comparable datasets and refined thresholds, the LRI could help practitioners assess

TABLE 4 Alignment between LRI national-scale indicators and selected Tanzanian policy commitments, SDG indicators, and restoration targets.

LRI national scale indicator	Relevant policy commitments, targets, and indicators	Baseline and target values
Water pillar		
Water quality and quantity	SDG VNR 6.3.2: Number of focused water quality monitoring networks established	Baseline: 430 in 2019
		Target: none found
	National Climate Change Response Strategy (2021–2026): Promote water resilience in all basins	Baseline: none found
		Target: Installation of integrated water resources management tools in all 9 basins by 2026
Access to quality water for different uses: drinking water, sanitation, irrigation	SDG VNR 1.4.1: Population living in households with access to drinking water and sanitation	Baseline: Drinking water 78.8% in 2022
		Target: none found
		Baseline: Sanitation 42.7% by 2022
		Target: none found
	SDG VNR 2.4.1: Area under irrigation	Baseline: 727,281 ha by 2022
		Target: none found
	Tanzania Nationally Determined Contribution 2021: Increase access to clean and safe water in rural areas by 2030	Baseline: 67.7% in 2015
		Target: 100% by 2030
Adaptive capacity pillar		
Agricultural productivity	SAGCOT Green Print 2012: Generate an additional 2.2 million tons of rice and field crops per year	Baseline none found
		Target: none found
Carbon stored in forest and agricultural land including associated emissions reductions	SAGCOT Green Print 2012: Carbon contained in these forests, combined with emissions reductions associated with AGG, will generate net emissions reductions of about three million tons of CO ² -equivalent per year by 2030	Baseline: none found
		Target: three million tons of CO ² -equivalent per year by 2030
	Land Degradation Neutrality Target: Increase soil organic carbon in cropland	Baseline: none found
		Target: 54.5tons/ha by 2030
Adoption of sustainable land management practices, certification and agroecological markets	Tanzania National Ecological Organic Agriculture Strategy 2023–2030: Number of farmers practicing Ecological Organic Agriculture (EOA)	Baseline: 1,000,000 in 2023
		Target: 3,000,000 Farmers received and gained knowledge on EOA Practices by 2030, led by Tanzania Organic Agriculture Movement
	National Climate Change Response Strategy (2021–2026): Improved number of ha of agricultural land	Baseline: None found
		Target: At least 50% of the community practicing climate smart for agricultural land management by 2026
	Changwon Initiative: Restore forests through sustainable forest management	Baseline: None found
		Target: 11,011,950 ha by 2030

(Continued)

TABLE 4 (Continued)

LRI national scale indicator	Relevant policy commitments, targets, and indicators	Baseline and target values
Healthy and diverse ecosystems	SDG VNR 15.1.1: Area of natural forest cover (Ha)	Baseline: 28,797,926 in 2017
		Target: none found
	Bonn Challenge: Restore 5.2 million hectares of land by 2030	Baseline: none found
		Target: 5.2 mn ha in restoration by 2030
		Baseline: none found
		Target: 1.5 million trees must be planted in each district each year until 2026
	National Biodiversity Strategy and Action Plan 2015–2020 Strategic goal D, target 14: “by 2020, ecosystems that provide essential services, related to water, and contribute to health, livelihoods and well-being, are restored and safeguarded, taking into account the needs of women, local and vulnerable communities.”	Baseline: none found
		Target: none found
Resilient communities to shocks, disasters and climate change	Tanzania Draft Development Vision 2050: A climate-resilience society that embraces climate-smart practices and targeted resilience building measures to adapt to the impacts of climate change	Baseline: None found
		Target: At least 3 million ha of degraded forest land restored by 2026
		Baseline: none found
Poverty rate	SDG VNR 1:2:1 Proportion of population living below the national poverty line	Baseline: none found
		Target: 0% poverty by 2030
	AGCOT/SAGCOT: Lift 2 million out of poverty	Baseline: 31.3% rural in 2017
		Target: 2 million by 2030

The table summarises examples of how LRI indicators correspond to national and international policy frameworks relevant to water security, sustainable land management, ecosystem restoration, climate resilience, agricultural productivity, and poverty reduction. Baseline and target values are reported where they were available in the reviewed policy documents, voluntary national review indicators, national strategies, or programme commitments. “None found” indicates that no explicit quantitative baseline or target value was identified in the reviewed source, rather than confirming that no such target exists.

whether interventions are contributing to measurable improvements in soil fertility, water access, sustainable land management adoption, productivity, land-cover condition, or other priority indicators. This repeat monitoring would also support adaptive management by identifying where course correction may be needed. For example, if soil nitrogen and phosphorus remain low despite soil fertility interventions, this may indicate the need to revise nutrient management approaches, while changes in water quality can help assess the effectiveness of upstream land and water management interventions (Kidane and Alemu, 2015; Issaka and Ashraf, 2017). In this way, the LRI can help stakeholders move from baseline assessment towards an iterative process of monitoring, learning, and management adjustment.

Finally, the Kilombero case study illustrates how the LRI can help connect landscape-level monitoring to policy and governance processes. By aligning selected indicators with policy commitments and reporting frameworks, the LRI can help bridge the gap between local data collection and higher-level decision-making, which is a recognised challenge in multi-scale environmental governance (Cash et al., 2006; Reed et al., 2016). Several indicators, including water access, water quality, sustainable land management, ecosystem restoration, agricultural productivity, climate resilience, and poverty reduction, correspond to national targets and reporting frameworks currently relevant in Tanzania (Table 4). This helps situate the Kilombero baseline findings within wider policy and programme objectives, while recognising that the case-study results represent landscape-level evidence rather than direct national reporting. Emerging interest from institutions such as AGCOT in Tanzania [CEO Roundtable of Tanzania (CEO RT), 2026] and the Monitoring, Reporting and Verification unit of the Ministry of Agriculture, Environment and Fisheries in Mozambique (A. Muhate, personal communication, 20 January 2026) further suggests the practical relevance of the LRI as a demand-driven monitoring tool. Such uptake could help strengthen data-to-decision pathways through developing the evidence base needed to attract investment, support collaboration in landscape approaches, reduce uncertainty, and inform the scaling of integrated landscape management and nature-based solutions (Chausson et al., 2020; Seddon et al., 2020; Reed et al., 2016).

5 Limitations

The LRI is designed to provide a broad, integrated picture of landscape resilience within individual landscapes, rather than a detailed diagnostic tool for any single ecosystem component. Differences in data provenance, data quality, use of proxy indicators, threshold-setting approaches, and context-specific sub-indicator selection may limit direct comparison of LRI scores across landscapes. As such, results should be interpreted with an understanding of the scope and representativeness of each indicator and sub-indicator, and not used to compare resilience across landscapes. Further, the LRI is not presented as a mechanistic model of resilience dynamics, but rather a pragmatic monitoring and decision support tool. The four-point categories are simplified interpretive thresholds designed to support baseline assessment, communication, and repeat monitoring. These categories should therefore be considered provisional and refined as additional evidence, stakeholder priorities, and monitoring data become available.

Some indicators rely on proxy measures. For example, bird species richness provides a useful indicator of broader biodiversity trends and

ecosystem health (Davis et al., 2024; Whelan et al., 2008) but does not capture the full complexity of species composition or ecosystem function. The use of proxy indicators reflects a pragmatic balance between ecological completeness and feasibility, which is a common requirement in applied landscape monitoring contexts (Carignan and Villard, 2002). Where more targeted management decisions are required, additional taxonomic or habitat-specific data may be necessary.

Moreover, the sub-indicators were selected to balance ecological relevance with data availability. Consequently, the specific sub-indicators applied under each indicator may differ between landscapes, reflecting differences in landscape characteristics, sampling objectives, and logistical constraints. While this flexibility enhances applicability across contexts, it may limit strict comparability between sites. However, this trade-off is necessary to ensure that indicators remain locally relevant and feasible to implement. Sub-indicator scores were equally weighted in the Kilombero baseline assessment as a transparent default approach, rather than as an assumption that all sub-indicators are equally important across all landscape contexts. The relative importance of individual sub-indicators may vary according to management objectives, dominant land uses, stakeholder priorities, policy targets, and data availability. For example, soil organic carbon in cropland may be weighted differently from soil organic carbon in woodland where agricultural production is the primary management concern. Future applications of the LRI may therefore apply stakeholder-agreed or objective-specific weights where these are developed through a transparent participatory process.

Where possible, scoring thresholds were derived from scientific literature and/or locally relevant policy standards. However, in cases where such benchmarks are not available, expert judgement was applied to define provisional thresholds. These should be considered adaptive and refined over time as additional evidence, policy targets or programme-specific targets become available.

As the Kilombero case study represents a baseline application of the LRI, the current scores should be interpreted as an initial baseline assessment of indicator condition rather than evidence of temporal change. Quantitative change will require repeat monitoring, where new sub-indicator values are compared against the same or refined threshold categories to assess whether scores improve, decline, or remain stable over time. Additionally, temporal dynamics also present important limitations. Many ecological processes respond slowly to interventions, meaning that detectable changes in indicators such as soil carbon or biomass may only emerge over longer timeframes (Smith, 2004). In addition, seasonal variability can significantly influence certain indicators, particularly those related to soil moisture and water quality (Daniel et al., 2017; Selemani et al., 2017). These factors should be considered when designing monitoring intervals and interpreting results.

6 Conclusion

This study highlights several important implications for the design and implementation of landscape resilience monitoring systems. First, there is a clear need for an integrated monitoring tool that moves beyond sector-specific approaches. Many existing systems track similar environmental and socio-economic outcomes but do so in isolation, limiting their ability to inform coordinated decision-making. The LRI differs from such approaches by integrating diverse indicators across land, water and adaptive capacity dimensions into a single tool that can

be used to identify where interventions, including NbS, can strengthen resilience across sectors. Second, balancing scientific robustness with practical feasibility is critical. While comprehensive datasets are desirable, monitoring systems must remain implementable within resource constraints. In contrast to more fixed monitoring frameworks, the LRI combines core indicators with flexible, context-specific sub-indicators, providing a practical pathway for applying landscape resilience monitoring across diverse landscapes. Third, strengthening the link between data collection, decision-making, and investment is essential. Monitoring systems are most effective when they directly inform planning, investment, and adaptive management. The LRI contributes to this by generating actionable insights at multiple scales, from farm-level decision-making to national reporting. In doing so, the LRI links local-level interventions to national policy priorities, thereby supporting investment in NbS and helping to catalyse systemic change in the way land is managed. Finally, the findings highlight the importance of institutional coordination and data sharing. Without mechanisms to integrate data across sectors and administrative levels, the value of monitoring systems remains limited. Embedding frameworks such as the LRI within existing policy and governance structures can help ensure that landscape-level data contributes to broader development and climate objectives. Rather than replacing existing monitoring systems, the LRI is intended to complement and strengthen them by acting as an integrated tool that links data across sectors and scales.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Author contributions

WO: Conceptualization, Supervision, Funding acquisition, Project administration, Methodology, Writing – original draft, Writing – review & editing, Data curation, Validation. ED: Funding acquisition, Validation, Methodology, Writing – original draft, Data curation, Writing – review & editing, Conceptualization. SO: Methodology, Writing – original draft, Data curation, Conceptualization. EI: Methodology, Conceptualization, Data curation, Writing – original draft. SD: Writing – review & editing, Data curation. MAB: Funding acquisition, Conceptualization, Writing – review & editing. MM: Writing – review & editing, Data curation. ME: Data curation, Writing – review & editing. CM: Writing – review & editing, Data curation. AM: Writing – review & editing, Data curation. AA-D: Writing – review & editing, Funding acquisition, Conceptualization.

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Conflict of interest

WO is employed by Terralytics Consulting Ltd. ED, SO, and EI were bona fide subcontractors of Terralytics Consulting Ltd.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fsufs.2026.1881795/full#supplementary-material>

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